

REG NO: 22TD0053 NAME: MOHAMED B SIRAJUDEEN

YEAR/SEM/SEC: Y3/S6/B

AIM:

SOURCE CODE:

;|||||;

;|||||;

```

;||      Title      : Buzzer.c

```

```

;||      Status      : Completed!
;||

```

```
;|| Features : ON/Off buzzer with time delay intervals ||;
```

```

;||      Oscillator      :12MHz
;||

```

```

;|| Enable Buzzer, Put Jumper JP1 to 'E' Mode. ||;

```

|||||;

```
//<<<<<<<<<<<<<<<<<< DIRECTIVES >>>>>>>>>>>>>>>>
```

```
#include <LPC214x.h>
```

```
#include <stdio.h>
```

```
#define BUZZ 7
```

```
void Delay(void);
```

```
void Wait(void);
```

```
void Delay()
{
    unsigned int i,j;
    for(i=0;i<1000;i++)
        for(j=0;j<700;j++);
}

void main()
{
    PINSEL0      =      0x00;           //Configure Port0.7 as GPIO
    IODIR0       =      3 << BUZZ;     //Configure Port0.7 as OutPut pin
    while(1)
    {
        IOSET0 = 1 << BUZZ;
        Delay();
        IOCLR0 = 1 << BUZZ;
        Delay();
    }
}
```

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YEAR/SEM/SEC: Y3/S6/B

AIM:

SOURCE CODE:

```
;/////////////////////////////////////////////////////////////////;
;/////////////////////////////////////////////////////////////////;

;|      Title           : LCD Demo.c                               |;
;|      Status          : Completed                                |;
;|      Features         : Program to Display Message in 2 lines   |;
;|      Description      : LCD Used :2x16 LCD in 4-Bit Mode        |;
;|      Control Lines    :P0.16 (RS) | P0.17 (RW) | P0.18 (EN)      |;
;|      Data Lines       :P0.19 - P0.22                             |;
;|      Enable LCD, turn on switch SW28 at LCD label mark position |;
;/////////////////////////////////////////////////////////////////;

#include <LPC214x.H>

#define RS      0x10000

#define RW      0x20000

#define EN      0x40000

void lcd_cmd (unsigned char);

void lcd_data (unsigned char);

void lcd_initialize (void);

void lcd_display (void);

void LCD4_Convert(unsigned char);

const unsigned char cmd[4] = {0x28,0x0c,0x06,0x01};           //lcd commands
```

```

unsigned char msg[] = {">PS-Primer 2148<"};           //msg
unsigned char msg1[] = {":: LCD Demo! ::"};          //msg1
void delay(unsigned int n)
{
    unsigned int i,j;
    for(i=0;i<n;i++)
    {
        for(j=0;j<12000;j++)
        {;}
    }
}

//-----
//          LCD Command Send
//-----

void lcd_cmd(unsigned char data)
{
    IOCLR0    |= RS;           //0x1000;           //RS
    IOCLR0    |= RW;           //0x2000;           //RW
    LCD4_Convert(data);
}

void lcd_initialize(void)
{
    int i;
    for(i=0;i<4;i++)
    {
        IOCLR0 = 0xF << 19;    //IOCLR 0/1
    }
}

```

```

        lcd_cmd(cmd[i]);

        delay(15);

    }

}

//-----

//          LCD Data Send

//-----

void lcd_data (unsigned char data)

{

    IOSET0      |= RS;                //0x1000;        //RS

    IOCLR0      |= RW;                //0x2000;        //RW

    LCD4_Convert(data);

}

//-----

//          LCD Display Msg

//-----

void lcd_display (void)

{

    char i;

    /* first line message */

    lcd_cmd(0x80);

    delay(15);

    i=0;

    while(msg[i]!='\0')

    {

        delay(5);

```

```

        lcd_data(msg[i]);

        i++;

    }

    delay(15);

    /* second line message */

    lcd_cmd(0xc0);

    delay(15);

    i=0;

    while(msg1[i]!='\0')
    {

        delay(5);

        lcd_data(msg1[i]);

        i++;

    }

    delay(15);
}

void LCD4_Convert(unsigned char c)
{

    if(c & 0x80) IOSET0 = 1 << 22; else IOCLR0 = 1 << 22;

    if(c & 0x40) IOSET0 = 1 << 21; else IOCLR0 = 1 << 21;

    if(c & 0x20) IOSET0 = 1 << 20; else IOCLR0 = 1 << 20;

    if(c & 0x10) IOSET0 = 1 << 19; else IOCLR0 = 1 << 19;


    IOSET0      = EN;
                //0x4000;                //EN

    delay(15);

```

```

        IOCLR0      = EN;
        //0x4000;          //EN

        if(c & 0x08) IOSET0 = 1 << 22; else IOCLR0 = 1 << 22;

        if(c & 0x04) IOSET0 = 1 << 21; else IOCLR0 = 1 << 21;

        if(c & 0x02) IOSET0 = 1 << 20; else IOCLR0 = 1 << 20;

        if(c & 0x01) IOSET0 = 1 << 19; else IOCLR0 = 1 << 19;

        IOSET0      = EN;
        //0x4000;          //EN

        delay(15);

        IOCLR0      = EN;
        //0x4000;          //EN
    }

    void main()

    {

        PINSEL1      =      0;
        //Configure P0.16 - P0.31 as GPIO

        IODIR0       =      0xFF << 16;    //Configure Pins P0.16 - P0.22 as Output Pins

        lcd_initialize();          //Initialize LCD!

        lcd_display();            //Display Message in LCD

        while(1);

    }

```

RESULT:

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YEAR/SEM/SEC: Y3/S6/B

AIM:

SOURCE CODE:

```
;|||||;|||||;|||||;|||||;|||||;|||||;|||||;|||||;|||||;|||||;||
;|||||;|||||;|||||;|||||;|||||;|||||;|||||;|||||;|||||;|||||;||
;||      Title           : LED Switch.c                                ||;
;||      Status          : Completed!                                  ||;
;||      Features        : Program to read switch status and displayed in point LEDs.   ||;
;||      Description     : Slide Switch    : P1.24 - P1.31              ||;
;||      LED Port       : P1.16 - P1.23                                   ||;
;||      Place jumper J4 ('E' Label position)                             ||;
;||                                                                    ||;
;|||||;|||||;|||||;|||||;|||||;|||||;|||||;|||||;|||||;|||||;
```

```

/*****
*****/

```

```
//<<<<<<<<<<<<<<<<<Code Begins Here>>>>>>>>>>>>>>>>>>
```

```
#include <LPC214x.h>
```

```
#include <stdio.h>
```

```
#define LED 16
```



```
#define Switch 24
```

```
/****** Declarations *****/
```

```
void Delay(int);
```

```
/******  
/
```

```
int main(void)
```

```
{
```

```
    unsigned char Status=1;
```

```
    PINSEL2    &=    0xFFFFFFFF3;    //Configure P1.16 - P1.31 as GPIO
```

```
    IO1DIR     =     0x00 << Switch;    //Configure P1.24 - P1.31 as Input
```

```
    IO1DIR     |=     0xFF << LED;    //Configure P1.16 - P1.23 as Output
```

```
    IO1PIN=     0;
```

```
    while(1)
```

```
    {
```

```
        Status = 1;
```

```
        IOSET1     =     0xFF << LED;
```

```
        Delay (10);Delay (10);
```

```
        IOCLR1     =     0xFF << LED;
```

```
        Delay (10);Delay (10);Delay (10);
```

```
        while (~Status)
```

```
        {
```

```
            Status = ((IO1PIN & (0xFF << Switch)) >> Switch);
```

```
        IO1PIN=      ((~Status) << LED);  
    }  
  
    }  
}
```

```
void Delay(int n)  
{  
    int p,q;  
  
    for(p=0;p<n;p++)  
    {  
        for(q=0;q<0x9990;q++);  
    }  
}
```

RESULT:

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YEAR/SEM/SEC: Y3/S6/B

AIM:

SOURCE CODE:

```
#include <lpc214x.h>                //Includes LPC2148 register definitions

#define SW3          1<<24
#define SW4          1<<25
#define SW5          1<<26


#define MAX_BUFFER_SIZE 16

#define DEVICE_ADDR 0xA0

#define BLK_0 0x00

#define BLK_1 0x02

#define MAX_BLOCK_SIZE 256


#define LED1_ON() IO1SET=(1<<16) //I2C write indicator
#define LED2_ON() IO1SET=(1<<17) //I2C read indicator
#define LED3_ON() IO1SET=(1<<18) //Comm failure indicator
#define LED4_ON() IO1SET=(1<<19) //Comm success indicator


#define LED1_OFF() IO1CLR=(1<<16)
#define LED2_OFF() IO1CLR=(1<<17)
#define LED3_OFF() IO1CLR=(1<<18)
```

```
#define LED4_OFF() IO1CLR=(1<<19)
```

```
#define Fosc      12000000
```

```
#define Fcclk     (Fosc * 5)
```

```
#define Fcco      (Fcclk * 4)
```

```
#define Fpclk     (Fcclk / 4)
```

```
void Delay(unsigned char j);
```

```
void Send_Start(void);
```

```
void Send_Stop(void);
```

```
unsigned char Send_I2C(unsigned char *Data,unsigned char Len);
```

```
unsigned char Read_I2C(unsigned char *Data,unsigned char Len);
```

```
unsigned char Page_Write(unsigned char BLOCK_NUMBER,unsigned char  
BLOCK_ADDR,unsigned char *str);
```

```
unsigned char Page_Read(unsigned int BLOCK_NUMBER,unsigned char BLOCK_ADDR);
```

```
unsigned char I2C_Status(unsigned char status_code);
```

```
unsigned char I2C_WR_Buf[MAX_BUFFER_SIZE]="ARM7 PRIMER BRD ";
```

```
unsigned char I2C_RD_Buf[MAX_BUFFER_SIZE];
```

```
unsigned char I2C_ER_Buf[MAX_BUFFER_SIZE]={0,0,0,0,0,0,0,0,0,0,0,0,0,0};
```

```
unsigned char Status=0;
```

```
unsigned char Status_Flag=0;
```

```
void Delay(unsigned char j)
```

```
{
```

```
    unsigned int i;
```

```
    for(;j>0;j--)
```

```
{  
    for(i=0; i<60000; i++);  
}  
}
```

```
void Send_Start()  
{  
    I2C1CONSET=0x20;  
}
```

```
void Send_Stop()  
{  
    I2C1CONSET=0x10;  
}
```

```
// This function sends sequential data to the EEPROM 24LC04  
// The buffer size for EEPROM 24LC04 is 16 bytes  
// The Len parameter should not exceed this value  
unsigned char Send_I2C(unsigned char *Data,unsigned char Len)  
{  
    while(Len)  
    {  
        I2C1DAT=*Data;  
        if(I2C_Status(0x28))  
        {  
            return 1;  
        }  
    }  
}
```

```

    }

    Len--;

    Data++;

}

return 0;

}

```

// This function reads random data from the EEPROM 24LC04

```

unsigned char Read_I2C(unsigned char *Data,unsigned char Len)

```

```

{
    while(Len)
    {
        if(Len==1) //Last byte
        {
            I2C1CONCLR=0x04; //set hardware to send nack

            if(I2C_Status(0x58)) //last byte has been received and NACK has been returned
            {
                return 1;
            }

            *Data=I2C1DAT;
        }
        else
        {
            I2C1CONSET=0x04; //set hardware to send ack

            if(I2C_Status(0x50)) //Byte has been received ACK has been returned

```

```
{  
    return 1;  
}  
*Data=I2C1DAT;  
}  
Data++;  
Len--;  
}  
return 0;  
}
```

```
unsigned char I2C_Status(unsigned char status_code)  
{  
    while(Status_Flag==0);  
    Status_Flag=0;  
    if(Status!=status_code)  
    {  
        return 1;  
    }  
    else  
    {  
        return 0;  
    }  
}
```

```

unsigned char Page_Write(unsigned char BLOCK_NUMBER,unsigned char BLOCK_ADDR,
unsigned char *str)

{

Send_Start();

if(I2C_Status(0x08)) //Start has been transmitted
{

return 1;

}


I2C1DAT=DEVICE_ADDR | BLOCK_NUMBER; // Send Address

if(I2C_Status(0x18)) //Device address, block num and write has
been transmitted

{

return 1;

}

I2C1DAT=BLOCK_ADDR; // Send block address

if(I2C_Status(0x28)) //Block address has been transmitted
{

return 1;

}


if(Send_I2C(str,MAX_BUFFER_SIZE)) //Send Data
{

Send_Stop();

return 1;

}

Send_Stop();

```



```
return 0;
```

```
}
```

```
unsigned char Page_Read(unsigned int BLOCK_NUMBER,unsigned char BLOCK_ADDR)
```

```
{
```

```
Send_Start();
```

```
if(I2C_Status(0x08)) //Start has been transmitted
```

```
{
```

```
return 1;
```

```
}
```

```
I2C1DAT=DEVICE_ADDR | BLOCK_NUMBER; // Send Address
```

```
if(I2C_Status(0x18)) //Device address, block num and write has been transmitted
```

```
{
```

```
return 1;
```

```
}
```

```
I2C1DAT=BLOCK_ADDR;
```

```
if(I2C_Status(0x28)) //Block address has been transmitted
```

```
{
```

```
return 1;
```

```
}
```

```
Send_Start(); // Repeat Start
```

```
if(I2C_Status(0x10)) //Repeated Start has been transmitted
```

```
{
```

```
return 1;
```

```
}
```

```
I2C1DAT=DEVICE_ADDR | BLOCK_NUMBER | 0x01; //Device address, block num and read  
has been transmitted
```

```
if(I2C_Status(0x40)) //
```

```
{
```

```
return 1;
```

```
}
```

```
if(Read_I2C(I2C_RD_Buf,MAX_BUFFER_SIZE)) //Receive 16bytes of Data from EEPROM
```

```
{
```

```
Send_Stop();
```

```
return 1;
```

```
}
```

```
Send_Stop();
```

```
return 0;
```

```
}
```

```
void __irq I2C1_Status(void)
```

```
{
```

```
Status_Flag=0xFF; //update status flag
```

```
Status=I2C1STAT; //Read Status byte
```

```
I2C1CONCLR=0x28;
```

```
VICVectAddr = 0x00; //Acknowledge Interrupt
```

```
}
```

```

void I2C_Init()

{

    PINSEL0|=0x30C00000;

    I2C1CONCLR=0x6C;

    I2C1CONSET=0x40;

    I2C1SCLH =80;

    I2C1SCLL =70;


    /* Init VIC for I2C1 */

    VICIntSelect = 0x00000000;    // Setting all interrupts as IRQ(Vectored)
    VICVectCntl0 = 0x20 | 19;    // Assigning Highest Priority Slot to I2C1 and enabling this slot
    VICVectAddr0 = (unsigned long)I2C1_Status; // Storing vector address of I2C1
    VICIntEnable = (1<<19);


}


void UART0_Init(unsigned int baudrate)

{

    unsigned int BAUD_VAL;

    BAUD_VAL = (Fpclk/(16*baudrate));


    PINSEL0 |= 0x00000005;    // Enable rx,tx

    U0LCR = 0x00000083;    // 8 bit data,1 stop bit,no parity bit

    U0DLL = BAUD_VAL & 0xFF; // U0DL for low byte

    U0DLM = (BAUD_VAL>>8); // U0DL for high byte

    U0LCR = 0x00000003;    // DLAB =0

```

```
}
```

```
void UART0_Txmt(unsigned char Chr) /* Write character to Serial Port */
```

```
{
```

```
    while (!(U0LSR & 0x20));
```

```
    U0THR = Chr;
```

```
}
```

```
unsigned char UART0_Rcv(void) /* Write character to Serial Port */
```

```
{
```

```
    while (!(U0LSR & 0x01));
```

```
    return (U0RBR);
```

```
}
```

```
void UART0_puts(unsigned char *string)
```

```
{
```

```
    while(*string)
```

```
        UART0_Txmt(*string++);
```

```
}
```

```
int main(void)
```

```
{
```

```
    PINSEL2 = 0;
```

```
    IO1DIR = (1<<19) | (1<<18) | (1<<17) | (1<<16); // Set P1.16, P1.17, P1.18, P1.19 as Output
```

```
    IO1DIR &= 0xC7ffff;
```

```
LED1_OFF();LED2_OFF();LED3_OFF();LED4_OFF();
```

```
UART0_Init(9600);
```

```
I2C_Init();
```

```
UART0_puts("\033[2J");
```

```
UART0_puts ("***** ARM Primer LPC-2148 I2C EEPROM Demo *****\n\n\r");
```

UART0_puts

[illegible]

```
UART0_puts ("[~] Turn SW 20 ON to Write default data to EEPROM! \n\r");
```

```
UART0_puts("[~] Turn SW 21 ON to Read and Display data from EEPROM! \n\r");
```

```
UART0_puts ("[~] Turn SW 22 ON to Erase data from EEPROM \n\r");
```

```
while(1)
```

$$\{$$

```
if ((IOPIN1 & SW3) == 0) /*...To Load the Default Data to the EEPROM ...*/
```

$$\{$$

```
LED1_ON();    //Write Indicator
```

```
if(Page_Write(BLK_1,0x00,I2C_WR_Buf))
```

$$\{$$

```
UART0_puts("Write Failed");
```

```
LED4_ON();
```

}

```
Delay(50);
```

```
LED1_OFF();
```

```
UART0_puts("Data Written Successfully\r\n");
```

```

        while (!(IOPIN1 & SW3));

    }

    else if ((IOPIN1 & SW4) == 0) /*..To Read the Data Stored in the EEPROM...*/
    {

        LED2_ON();    //Read indicator

        if(Page_Read(BLK_1,0x00))
        {

            UART0_puts("Read Failed");

            LED4_ON();

        }

        else

        {

            UART0_puts("Data Read:");

            UART0_puts(I2C_RD_Buf);

            UART0_puts("\r\n");

        }

        Delay(50);

        LED2_OFF();

        while (!(IOPIN1 & SW4));

    }

    else if ((IOPIN1 & SW5) == 0)

    {

        LED3_ON();    //Write Indicator

        if(Page_Write(BLK_1,0x00,I2C_ER_Buf))

        {

            UART0_puts("Write Failed");

```

```
        LED4_ON();  
    }  
    Delay(50);  
    LED3_OFF();  
    UART0_puts("Data Erase Successfully\r\n");  
    while (!(IOPIN1 & SW5));  
}  
  
}  
  
}
```

RESULT:

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YEAR/SEM/SEC: Y3/S6/B

AIM:

SOURCE CODE:

;

;|||||;

```

;|| Title : Ex13 UART0 Demonstration ||;

```

```

;||      Status      : Completed!                                ||;

```

```

;|| Features : Program to put and get characters to and from PC Hyper terminal ||;

```

```

;|| window at 9600 baud rate. ||;

```

;		Description	:	UART0 Pins	:	P0.0 - TxD			;
---	--	-------------	---	------------	---	------------	--	--	---

|| : P0.1 - RxD ||;

|| Connect Serial Cable at P1(Board DB9 connector) to PC's DB9 Connector ||

;

```
#define CR    0x0D
```

```
#include <LPC21xx.H>
```

```
void init_serial (void);
```

```
int putchar (int ch);
```

```
int getchar (void);
```

```
unsigned char test;
```

```
//<<<<<<<<<<<<<<<Code Begins Here>>>>>>>>>>>>>>>
```


[illegible]

```
//<<<<<<<<<<<<<< Puthar Function >>>>>>>>>>>>>>>>
```

```
int putchar (int ch)    /* Write character to Serial Port */
```

 $\{$

```
if (ch == '\n') {
```

```
while (!(U0LSR & 0x20));
```

```
U0THR = CR;          /* output CR */
```

}

```
while (!(U0LSR & 0x20));
```

```
return (U0THR = ch);
```

$$\}$$

```
//<<<<<<<<<<<<<< Getchar Function >>>>>>>>>>>>>>>>
```

```
int getchar(void) /* Read character from Serial Port */
```

$$\{$$

```
while (!(U0LSR & 0x01));
```

```
return (U0RBR);
```

$$\}$$

RESULT:

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AIM:

SOURCE CODE:

```

;|||||;
;|||||;

;|| Title : Ex5 External Interrupt Study Code ||;
;|| Status : Completed! ||;
;|| Features : External Interrupts are processed and the number of press is ||;
;|| displayed in UART0 ||;
;|| Description : INT 1 (P0.14) -Press to Increment the data ||;
;|| INT 2 (P0.15) - Press to Increment the data ||;
;|| UART0 Pins : TxD : P0.0 | RxD : P0.1 ||;
;|| UART0 Configured for 9600 bps @ 15 MHz PCLK ||;
;|| output show by hyperterminal ||;
;|||||;

```

```

/*****
*****/

```

```
//<<<<<<<<<<<<<<<<<<<Code Begins Here>>>>>>>>>>>>>>>>>>
```

```
#include <lpc214x.h>
```

```
#include <stdio.h>
```

```
int volatile EINT1      =      0;
```

```
int volatile EINT2      =      0;
```

```
void ExtInt_Serve1(void)__irq;
```

```
void ExtInt_Serve2(void)__irq;
```

```
/*-----<  INT2 Initialization  >-----*/
```

```
void ExtInt_Init2(void)
```

```
{
```

```
    EXTMODE |= 4;
```

```
    //Edge sensitive mode on EINT2
```

```
    EXTPOLAR = 0;
```

```
    //Falling Edge Sensitive
```

```
    PINSEL0 |= 0x80000000;
```

```
    //Enable EINT2 on P0.15
```

```
    VICVectCntl1 = 0x20 | 16;    //Use VIC1 for EINT2 ; 16 is index of EINT2
```

```
    VICVectAddr1 = (unsigned long) ExtInt_Serve2;//Set Interrupt Vec Addr in VIC1
```

```
    VICIntEnable |= 1<<16;//Enable EINT2
```

```
}
```

```
/*-----<  INT1 Initialization  >-----*/
```

```
void ExtInt_Init1(void)
```

```
{
```

```
    EXTMODE |= 2;
```

```
    //Edge sensitive mode on EINT1
```

```

    EXTPOLAR = 0;
    //Falling Edge Sensitive

    PINSEL0 |= 0x20000000;
    //Enable EINT2 on P0.14

    VICVectCntl0 = 0x20 | 15;    //Use VIC0 for EINT2 ; 15 is index of EINT1

    VICVectAddr0 = (unsigned long) ExtInt_Serve1; //Set Interrupt Vec Addr in VIC0

    VICIntEnable |= 1<<15; //Enable EINT1
}

/*-----< Serial Initialization >-----*/

void Serial_Init(void)
{
    PINSEL0 |= 0X00000005;    //Enable Txd0 and Rxd0

    U0LCR = 0x00000083;    //8-bit data, no parity, 1-stop bit

    U0DLL = 0x00000061;    //XTAL = 12MHz (pclk = 60 MHz, VPB =
15MHz(pclk=cclk/4)

    U0LCR = 0x00000003;    //DLAB = 0;
}

/*-----< Delay Function >-----*/

void DelayMs(unsigned int count)    {

    unsigned int i,j;

    for(i=0;i<count;i++)

        {

            for(j=0;j<1000;j++);

        }

}

```

```
/*-----< Main Function >-----*/
```

```
void main(void)
```

```
{
```

```
    Serial_Init();
```

```
    ExtInt_Init1();           //Initialize Interrupt 1
```

```
    ExtInt_Init2();           //Initialize Interrupt 2
```

```
    putchar(0x0C);
```

```
    printf ("***** External Interrupt Demo  
*****\n\n\r");
```

```
    DelayMs(100);
```

```
    printf (">>> Press Interrupt Keys SW2(INT1) and SW3(INT2) for Output... \n\n\r");
```

```
    DelayMs(100);
```

```
    while(1)
```

```
    {
```

```
        DelayMs(500);
```

```
        printf("INT1 Generated : %d | INT2 Generated : %d \r", EINT1, EINT2);
```

```
        DelayMs(500);
```

```
    }
```

```
}
```

```
/*-----< Interrupt Sub-Routine >-----*/
```

```
void ExtInt_Serve1(void)__irq
```

```
{
```

```
    ++EINT1;
```

```
        EXTINT |= 2;

        VICVectAddr = 1;
    }

void ExtInt_Serve2(void)__irq
{
    ++EINT2;

    EXTINT |= 4;

    VICVectAddr = 0;
}
```

RESULT: